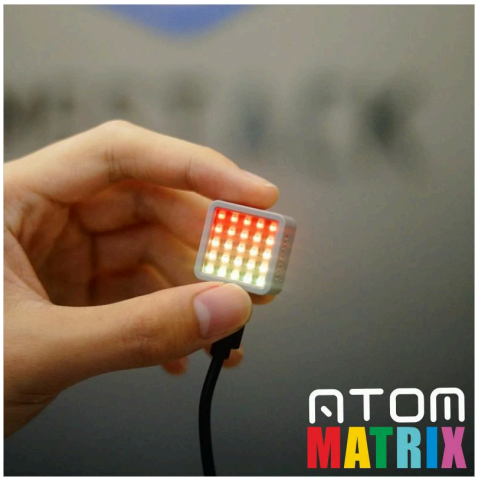
Atom-Matrix

SKU:C008-B



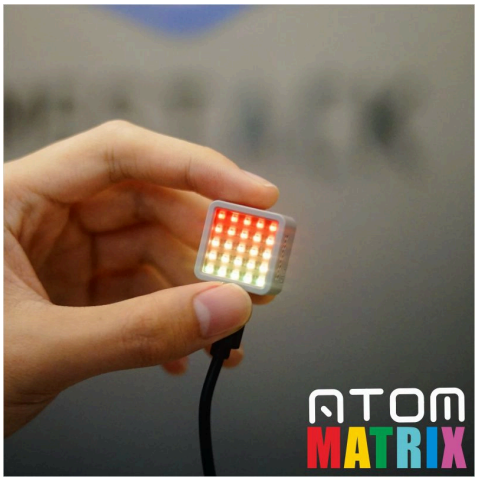


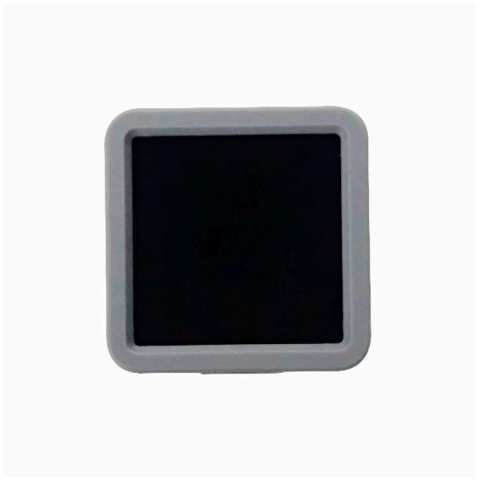














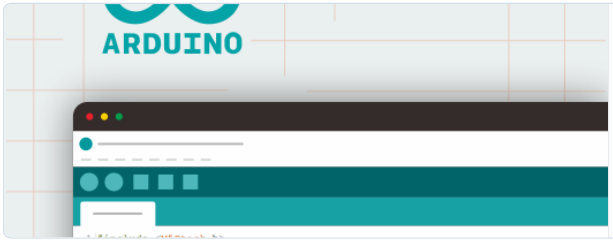
Description

Atom-Matrix is an ultra-compact development board in the M5Stack series, measuring only **24.0 x 24.0 mm**. It offers additional GPIOs for user customization and is ideal for embedded smart-hardware projects. Powered by the **ESP32-PICO-D4** with an integrated **Wi-Fi** module and **4 MB** SPI flash, the board carries an IR transmitter, a **5 × 5 RGB LED matrix**, a built-in **IMU (MPU6886)**, and a programmable button hidden beneath the Neo-LED matrix. A USB Type-C port enables quick program upload/download, and an HY2.0-4P connector is provided for peripheral expansion. An M2 mounting hole on the back allows secure installation.

Note

When using the FastLED library, a recommended brightness value for the RGB LEDs is 20. Do not set the brightness too high, or the LEDs and acrylic panel may be damaged. (In the Atom-Matrix library, we have mapped a suitable brightness range of 0–100.) **Excessive brightness or full-on operation can generate high temperatures and damage the enclosure.**

Tutorial



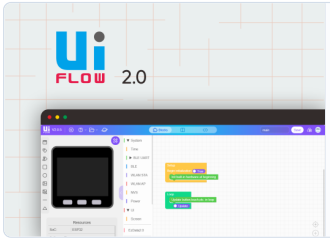
Arduino IDE

This tutorial explains how to program and control the Atom-Matrix with the Arduino IDE.



UiFlow1

This tutorial explains how to control the Atom-Matrix with the UiFlow1 graphical programming platform.



UiFlow2

This tutorial explains how to control the Atom-Matrix with the UiFlow2 graphical programming platform.

Features

- ESP32-based development
- Ultra-compact body
- Built-in 3-axis gyroscope and 3-axis accelerometer (I2C Address: 0x68)
- Programmable button
- RGB LED matrix display
- Infrared transmission
- Expandable pins and interfaces
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x Atom-Matrix

Applications

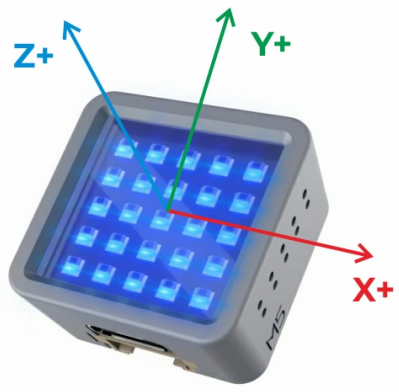
- IoT node
- Micro-controller

Specifications

Specification	Parameter
SoC	ESP32-PICO-D4 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency up to 240MHz
DMIPS	600
SRAM	520KB
Flash	4MB
Wi-Fi	2.4 GHz Wi-Fi
Input Voltage	5 V @ 500 mA
Host Interface	USB Type-C × 1, GROVE (I2C + I/O + UART) × 1
PINs	G19, G21, G22, G23, G25, G33
RGB LED	WS2812C 2020 × 25
MEMS	MPU6886 (I2C Address: 0x68)
IR	Infrared transmission
Button	Programmable button × 1
Antenna	2.4 G 3D antenna
Working Temp.	0 ~ 60 °C
Enclosure	Plastic (PC)
Product Size	24.0 x 24.0 x 13.8mm
Product Weight	7.3 g
Package Size	85.0 x 66.0 x 15.0mm
Gross Weight	12.6 g

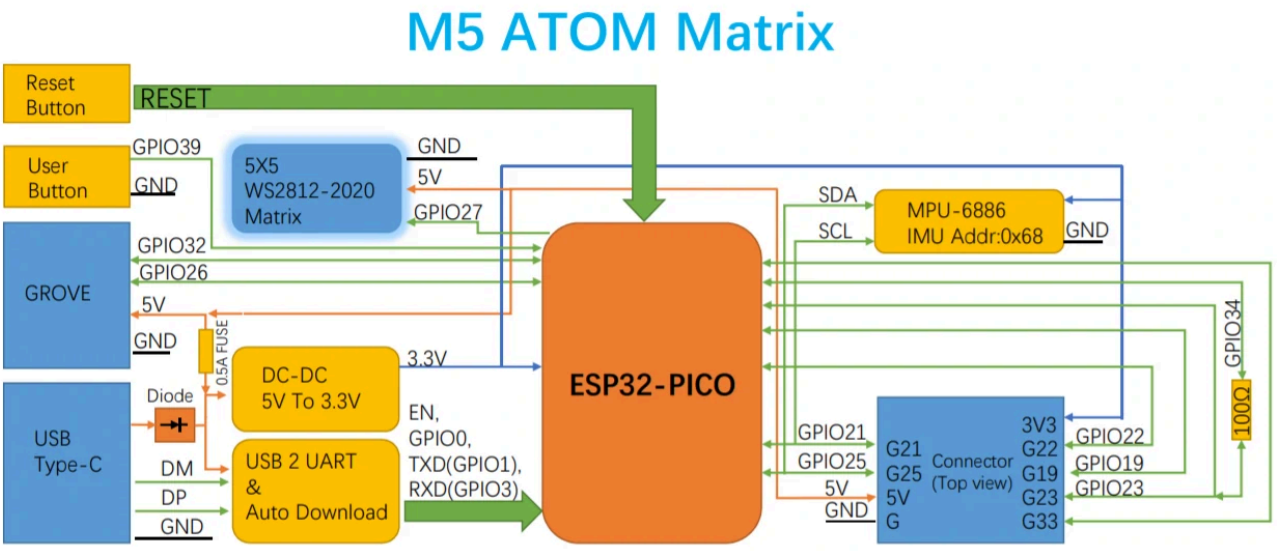
Operation Instructions

IMU Triaxial Direction Schematic Diagram

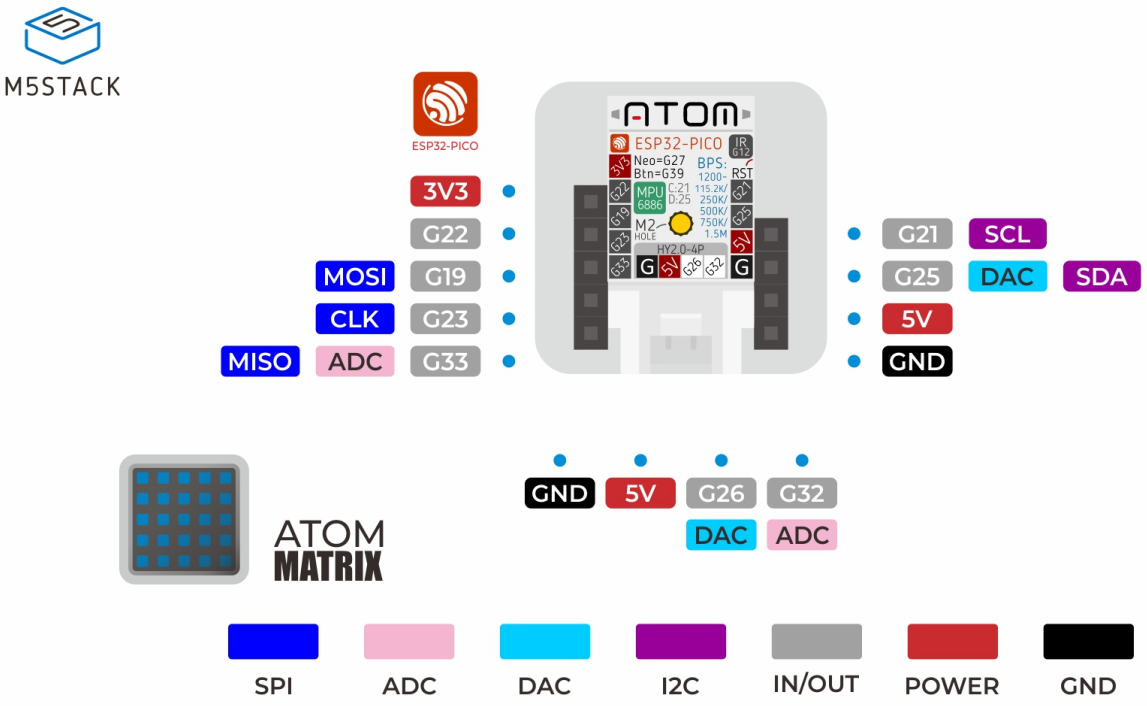


IMU ORIENTATION OF AXES

Schematics



PinMap



RGB & BUTTON & IR & MPU6886

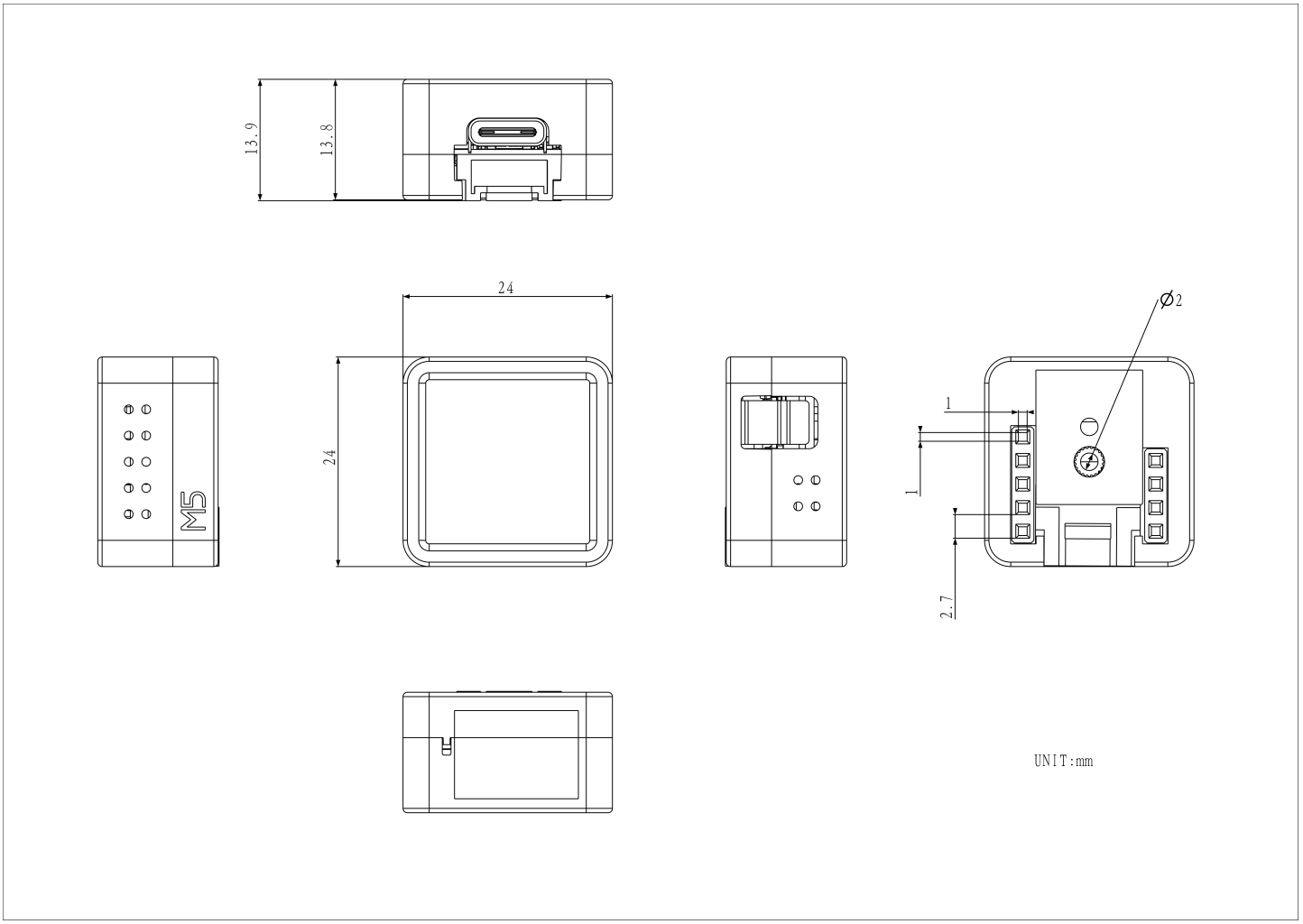
ESP32-PICO-D4	G27	G39	G12	G21	G25
RGB LED	RGB				
Button		Button			
IR			IR_TX		
MPU6886 (0x68)				SCL	SDA

Grove

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G26	G32

Model Size

- Atom-Matrix Model Size PDF



Structure

- Atom-Matrix Structure Files

Datasheets

- ESP32-PICO
- MPU6886
- WS2812C-2020

Softwares

Arduino

- Atom-Matrix Arduino Quick Start
- Atom-Matrix Arduino Library

UiFlow1

- Atom-Matrix UiFlow1 Quick Start

UiFlow2

- Atom-Matrix UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-atom]
platform = espressif32@6.7.0
board = m5stack-atom
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

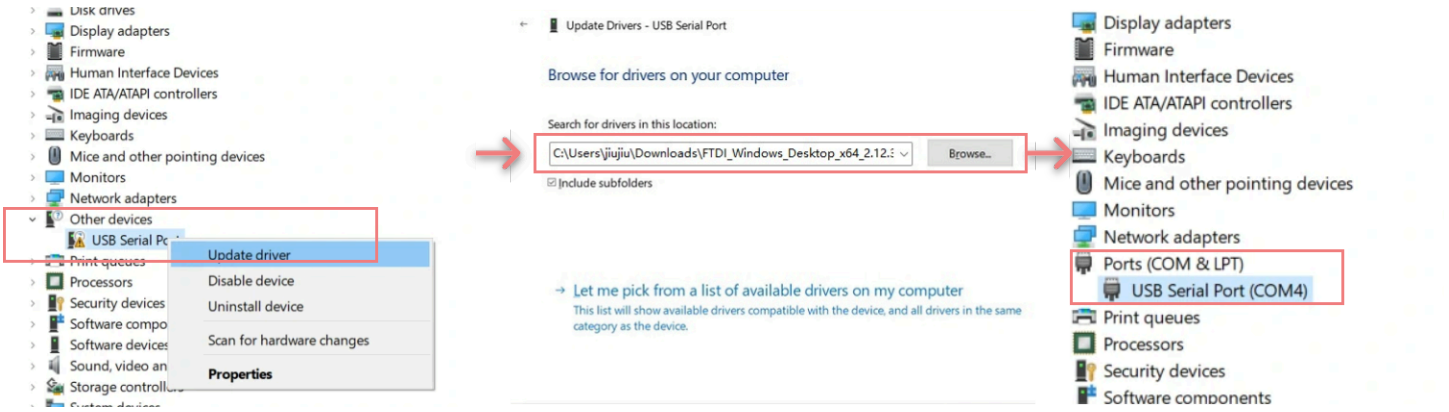
USB Driver

Atom-Matrix may not work without a driver on some systems. You can resolve this issue by manually installing the FTDI VCP driver.

1. Select the driver file that matches your operating system from the list below, download it, and extract it. You can also visit the FTDI Chip official website for additional FTDI driver versions.

Driver Name	Download Link
Windows_Desktop_x64	Download
Windows_Desktop_x86_32	Download
Windows_Desktop_ARM	Download
MacOS_Intel	Download
MacOS_ARM64	Download

2. Connect the device to your PC, open Device Manager, and install the FTDI driver (the screenshot below uses Windows 10 as an example). (Note: On some systems, you may need to install the driver twice for it to take effect. The unrecognized device is usually named `M5Stack` or `USB Serial`.)



Easyloader

Easyloader	Download	Note
Atom-Matrix Factory Test Easyloader	download	/

Video

Product Comparison

To compare information on the Atom series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.